MILITARY SPECIFICATION

NICKEL ALLOY, BARS, AND FORGINGS, 1,200° to 1,500°F OPERATING TEMPERATURES

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.- This specification covers a nickel-chromium-iron alloy similar to an alloy commercially designated as "Inconel X," in the form of bars, forging stock, or forgings.

1.2 Classification.- Materials shall be of one composition only and in the following conditions, forms, thermal treatments, and finishes (surface conditions), as specified. (See paragraph 6.1.)

Conditions

A - Hot-finished (rolled, forged, or extruded).
B - Forging quality (for subsequent hot manipulation).
C - Solution-treated (see paragraph 3.3.1).
D - Solution-treated and high-temperature aged (see paragraph 3.3.2).
E - Fully heat-treated (solution-treated, high-temperature aged, and aged) (see paragraph 3.3.3).

Finishes	Applicable to
(1) Rough centerless ground	Rounds, 3/8 to 1 in. dia excl.
(2) Rough-turned	Rounds, 1 in. dia and over.
(3) As hot-finished	Sections other than rounds, (limited sizes).

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specification and standard, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-M-151 Metals; General Specification for Inspection of

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STANDARDS

MIL-STD-129 Marking of Shipments

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be of such quality that the finished product shall comply with all requirements of this specification.

3.2 Chemical composition.- Chemical composition shall be as specified in table I.

TABLE I

Chemical composition

Element	Composition percent
Carbon (max)	0.08
Manganese	0.30 - 1.00
Silicon (max)	0.50
Sulfur (max)	0.01
Chromium	14.00 - 16.00
Nickel (min) 1/	70.0
Columbium 2/	0.70 - 1.20
Titanium	2.25 - 2.75
Aluminum	0.40 - 1.00
Iron	5.00 - 9.00
Copper (max)	0.20

1/ Including cobalt.
2/ Columbium plus tantalum.

3.3 Thermal treatment.-

3.3.1 Solution treatment (condition C).- Solution treatment shall consist of heating at 2,100° ±25°F, and holding at that temperature for 2 to 4 hours, followed by air cooling. This treatment shall be applicable to bars and forgings, except that, in the case of forgings, cooling by means other than air cooling may be employed when so stipulated in the contract or order, or as otherwise agreed upon between forging supplier and purchaser.

3.3.2 Solution treatment plus high-temperature aging (condition D).- Thermal treatment shall consist of solution treatment in accordance with paragraph 3.3.1, followed by heating to 1,550° ±25°F and holding at that temperature for 24 hours.

3.3.3 Full heat treatment (condition E).- Full heat treatment shall consist of solution treatment and high-temperature aging in accordance with paragraph 3.3.2, followed by heating to 1,300° ±25°F and holding at that temperature for 24 hours, followed by air cooling. Where convenient, air cooling after the 1,550°F aging treatment of paragraph 3.3.2 may be omitted and the furnace cooled to 1,300°F for the final aging treatment.

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3.4 Physical properties.-

3.4.1 Tensile properties.- Unless otherwise specified, there are no room-temperature tensile requirements for this material.

3.4.2 Room-temperature hardness.-

3.4.2.1 Materials ordered to condition E shall possess a room-temperature hardness of not less than Brinnell 262. Materials ordered to other conditions shall develop a hardness of not less than Brinnell 262 after receiving thermal treatment to condition E.

3.4.2.2 Room-temperature hardness requirements for forgings shall be as specified in the contract or order.

3.4.3 Stress-rupture properties.- After receiving heat treatment in accordance with paragraph 3.3.3, materials shall meet the following requirement.

3.4.3.1 A tensile test specimen, maintained at a temperature of 1,350° ±5°F while an axial load of 45,000 psi is applied continuously, shall not rupture in less than 100 hours. The test shall be continued, after the 100 hours, until the specimen ruptures, with the load maintained at 45,000 psi or increased to not more than 70,000 psi, as necessary to produce rupture. In either case, the elongation after rupture shall be measured at room temperature, using a gage length of four times the diameter of the specimen.

3.5 Tolerances.-

3.5.1 Diameter, thickness, or width.- The permissible variations in dimensions of bars shall be as shown in table II.

TABLE II

Permissible variations for hot-finished bars 1/

Dimensions 2/ Inch	Condition	Finish	Permissible variations, Inch	
			Plus	Minus
Rounds				
3/8 to 1, excl.	A, B 3/	(1)	0.016	0.016
1 and over	A, B 3/	(2)	0.031	0
Sections other than rounds				
1 and under	A	(3)	0.016	0.016
over 1 to 2, incl	A	(3)	0.031	0.016
over 2 to 4, incl	A	(3)	0.047	0.031
over 4	A	(3)	0.125	0.063

1/ Tolerances for machined sections, other than rough-turned rounds, finish (2), shall be as agreed upon between the manufacturer and the purchaser.

2/ Dimensions apply to diameter of rounds, to distance across flats of hexagons and squares, and separately to width and thickness of rectangles.

3/ Condition B (forging quality) may be spot ground to clean up minor surface imperfections, such spotted areas not to be deeper than 3 percent of the diameter.

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3.5.2 The permissible variations in dimensions of forgings shall be as specified in the contract or order.

3.5.3 Random lengths.- When cut lengths are not specified in the order, bars will be accepted in random mill lengths of 6 to 24 feet for sections weighing 25 pounds per foot or less. For sections weighing more than 25 pounds per foot, shorter lengths will be acceptable.

3.5.4 Cut lengths.- Bars of all sizes may be ordered to cut lengths or in lengths expressed as a multiple of a definite unit, with tolerances of +1/8 -0 inch. When multiple lengths are specified, 1/4-inch allowance per cut will be furnished, unless otherwise specified.

3.6 Identification of product.-

3.6.1 Bars 1/2 inch, or more, in diameter or 3/8 inch, or more, in width of flat face.- When specified, or for direct procurement by the Government, bars 3/8 inch, or more, in width of flat or 1/2 inch, or more, in diameter of round shall be legibly marked along the entire length with a row of recurring symbols. The symbols shall be repeated at intervals not greater than 3 feet. The marking shall not be affected by water, or rub off or be smeared by contact incident to normal handling during shipment or storage. The following information shall be shown:

MIL-M-8550
Physical condition
Surface condition
Manufacturer's name or trade-mark

3.6.2 Bars less than 3/8 inch in width of flat face or 1/2 inch in diameter.- Bars shall be securely bundled and tagged with the above information, using an oilproof tag. In addition, one tag, suitably marked, shall be placed near the middle on the inside of the bundle and securely fastened thereto.

3.7 Workmanship.- Bars, forging stock, and forgings shall be sound, clean, commercially straight, of uniform quality and condition, free from foreign material, pipes, slivers, and free from injurious defects such as laps, cracks, twists, seams, scale, or damaged ends or edges which detrimentally affect the suitability of the material for the service intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required herein for the testing of nickel-alloy bars and forgings are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section.

4.2 - Examination of product.- Bars and forgings shall be carefully examined as necessary to assure compliance with this specification with respect to workmanship, tolerances, and marking requirements.

4.3 Chemical analysis.-

4.3.1 Sampling.- Samples for check chemical analysis shall be selected by the Inspector in accordance with paragraph 4.3.1.1 of this specification and as described in Specification QQ-M-151, to represent each heat from which material is submitted for inspection. Samples shall be free from dirt, grit, and other foreign material, and shall weigh not less than 2 ounces.

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4.3.1.1 Samples shall be taken parallel to the axis of the bar and forging selected, from a zone midway between the center and the surface, except that material less than 1-1/4 inches thick shall be sampled through the entire cross section. Where material cannot be identified by heat, samples shall be taken from each 2,000 pounds submitted for acceptance at one time. Samples shall consist of not less than 2 ounces of chips or drillings.

4.3.1.2 Waiver.- Samples for check chemical analysis may be waived at the discretion of the Inspector, provided that all the material under inspection can be identified as being made from a heat previously analyzed and found to be in conformance with the chemical composition specified in table I.

4.3.2 Method.- Samples shall be prepared in accordance with the requirements of Specification QQ-M-151 and analyzed by wet chemical or spectrographic methods. In the event of dispute, analysis shall be by wet-chemical methods.

4.4 Hardness.-

4.4.1 Sampling (bars only).- One sample shall be selected from each heat ordered to condition A and fully heat-treated. In addition, one sample shall be selected from each heat for all conditions and thermal treatments of bars, other than fully heat-treated, such samples being selected either from a hot-finished bar or from a test ingot that has been given a minimum hot reduction of 50 percent.

4.4.2 Sampling (forgings).- Samples shall be selected as specified in the contract or order.

4.4.3 Preparation of specimens.- Samples of materials not in the fully heat-treated condition shall be heat-treated in accordance with paragraph 3.3.3 prior to testing. Test specimens shall be prepared as described in Specification QQ-M-151, unless otherwise specified by the contract or order.

4.4.4 Method.- Hardness determinations on bars shall be taken on flat specimens at midradius, unless otherwise specified. In the case of forgings, hardness determinations shall be taken as described in the contract or order.

4.5 Rejection.- In the event a lot of material fails to comply with all requirements of this specification, the lot shall be rejected, except that where failure may be due to inadequate heat treatment, the material may be reheat-treated and resubmitted for test. Only one such reheat treatment will be permitted.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packing.- All material shall be properly separated or segregated by size (section) and condition when packed for shipment.

5.2.1 Domestic packing.- Material shall be bundled, or otherwise prepared for shipment in units convenient to handle at destination. However, where truckload or carload quantities are involved, bundling may be omitted at the discretion of the procuring activity provided loading is done in a manner consistent with good shipping practice. Containers shall be so constructed and materials loaded in a manner to protect against surface injury and to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

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5.2.2 Overseas packing.- All material shall be securely crated, bundled, or boxe The gross weight of individual shipping containers shall not exceed approximately, 1,000 pounds.

5.3 Marking of shipments.- All packages and shipping containers shall be marked accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Nickel Alle Bars (or forgings, as appropriate), Nickel-Chrome-Iron, Specification MIL-N-8550; Finish and physical condition, Heat No., Method of manufacture (rolled or extruded, as appropria Shape and dimensions, Length.

5.3.1 Bundles.- If not boxed or crated, individual pieces or bundles shall have attached a metal tag impression stamped with the above information.

6. NOTES

6.1 Intended use.- The material is intended primarily for turbine parts requiring oxidation resistance and high strength at temperatures of 1,200° to 1,500°F.

6.1.1 Bars in physical conditions C and D are intended for parts to be rough machined by the user, and subsequently given the high-temperature aging and the final age hardening at 1,300°F, as applicable, followed by finish machining or grinding. The selection of materials in conditions C or D will depend on the type of machining and sequence of operations involved.

6.1.2 Bars in physical condition E are intended for parts that are to be fully heat-treated by the manufacturer and subsequently finish-machined. Where surface finish, rather than tool life, feeds and speeds, is significant, such material is often preferred.

6.1.3 Condition B (forging quality) rounds are intended for hot manipulation and subsequent thermal treatment and fabrication.

6.2 Ordering data.- Requisitions, contracts, and orders should state the following information:

(a) Size (dia of rounds or section size)
(b) Condition
(c) Thermal treatment
(d) Finish (surface condition)
(e) Random, cut, or multiple lengths. When multiple lengths are ordered, 1/4 inch per cut will be allowed, unless otherwise stipulated. When this allowance is not sufficient, the desired allowance per cut should be stipulated
(f) Quantity

6.3 Miscellaneous notes.-

6.3.1 Hardness.- The hardness of bars, given the various thermal treatments, will vary depending on the treatment used, the size of the section and the rapidity with which the material is cooled through the age hardening zone. In general, small sections that can be cooled more rapidly will be softer than large sections. The following hardness values, except for treatment E, are offered for information only and are not to be construed as the basis for acceptance or rejection:

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	Brinell hardness
Hot-finished (air-cooled)	225-300
Condition C (solution-treated)	140-280
Condition D (solution-treated and high-temperature aged)	200-280
Condition E (fully heat-treated)	1/ 262-340

1/ Required (see table II.)

6.3.2 Bars, other than rounds furnished to this specification, are available in limited sizes. The manufacturer should be consulted for recommended machining allowances.

6.3.3 Procurability.- Unless otherwise agreed upon, bars and forgings are procurable in the following combinations of conditions and finishes.

Bars	Conditions	Finishes 1/ (surface condition)
Rounds, 3/8 to 1 inch, dia, excl	A, C, D, E	(1)
Rounds, 1 inch dia and over	A, C, D, E	(2)
Sections other than rounds	A, C, D, E	(3)
Rounds, 3/8 to 1 inch dia, excl	B	(1)
Rounds, 1 inch dia and over	B	(2)

1/ Where material is ordered in physical conditions C, D, or E, the material shall be furnished with the oxide resulting from such thermal treatments.

6.3.3.1 Forgings are procurable with or without thermal treatment and with surface conditions (as forged, machined, or ground), as specified in the contract or drawing, or as otherwise agreed upon.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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